Subhajit Sarkar

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EDUCATION

National Institute of Technology, Silchar — **Assam, India**
B.Tech in Computer Science and Engineering; **CGPA: 7.73** — Aug 2024 - Jun 2028

- **Relevant Coursework:** Data Structures & Algorithms, OOP, Database Management Systems, Operating Systems, Computer Networks, DAA, Computer Architecture

TECHNICAL SKILLS

Languages & Frontend: C, C++, Python, JavaScript, SQL, React.js

Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication, Swagger/OpenAPI, Prisma ORM

AI & Systems: Agentic RAG, LLMs, LangGraph, OCR, Embeddings, ChromaDB, Docling, PyTorch, Triton, CUDA

Databases: PostgreSQL, MongoDB, MySQL

Core Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Distributed Systems

Tools: Git, GitHub, Docker, Postman, Render, Vercel

PROJECTS

PagedServe | *Python, PyTorch, Triton, CUDA, FastAPI* — Benchmarks GitHub · **Oct 2026**

- Architected a VRAM-efficient LLM serving engine for **4 GB consumer GPUs** using **continuous batching** and a block-based **paged KV cache**, serving Qwen2.5-0.5B within **∼25% of device memory** (about 1 GiB peak) while keeping KV-cache fragmentation **below 1%**.
- Improved aggregate throughput by **51%** at 8 concurrent requests through iteration-level scheduling, with **under 2%** additional VRAM; exposed request, status, and cancellation endpoints via **FastAPI** and validated the system with **128 automated tests**.
- Accelerated decode-phase attention with a GPU-optimized paged-attention backend (**Triton**), cutting kernel latency by **97%** and delivering **134% higher** end-to-end throughput with **57% lower** request latency at 8 concurrent requests on the same 4 GB device.

API Monitoring Platform | *Node.js, Express.js, PostgreSQL, Prisma ORM, Redis, JWT* — GitHub **June 2026**

- Built a SaaS-style monitoring backend with **31 REST endpoints** and a **7-model PostgreSQL schema**, tracking uptime, response time, and incidents; documented via **Swagger** and containerized with Docker.
- Automated **500+ scheduled health checks** with auto-opened and auto-resolved incidents, **email alerts**, and a nightly job purging data older than **30 days**.
- Secured with **JWT, Google OAuth, and OTP verification**, plus **rate limiting** of 5 auth attempts and 50 API requests per 15 minutes.

Agentic RAG Doc Analyzer | *Python, LangGraph, Gemini, Docling, RapidOCR, ChromaDB* — GitHub **March 2026**

- Built an Agentic RAG analyzer with **9+ specialized tools**, conversational memory, and multi-document management.
- Built a pipeline using **Docling, RapidOCR, validation, chunking, and embeddings** for content extraction.
- Orchestrated **LangGraph** workflows for semantic retrieval, information extraction, comparison, summarization, verification, and reasoning.

PROFILES & CERTIFICATIONS

- **Codeforces (Pupil):** Max rating **1274**. Profile
- **LeetCode (Knight):** Max rating **1888**; solved **600+** problems; top **5.05%** of contest participants. Profile
- **CodeChef (2 Star):** Max rating **1528**. Profile
- **AlgoTech Fellowship (ATF):** Qualified as Stage 2 candidate; ranked in the top **8%** nationwide among **250,000+** candidates. Credential
- **Full Stack Web Development Certification (Udemy).** Credential

POSITION OF RESPONSIBILITY

Computer Science Society, NIT Silchar — **Onsite**
Dev-Wing Member (Full-Time) — Sep 2025 - Present

- Collaborated in a development team to build and maintain web applications, improving functionality and UX.
- Contributed to feature development, debugging, testing, and code reviews, ensuring scalable, maintainable code.